Industrial Safety Processes

Metal Welding & Joining

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Types of Hot Metal Joining

- **Arc welding** (21 sub-types)
- **Oxyfuel gas welding** (4 sub-types)
- **Solid state welding** (13 sub-types)
- **Resistance welding** (11 sub-types)
- **Soldering** (9 sub-types)
- **Brazing** (10 sub-types)
- **Other** (19 sub-types)

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Flux

- Derived from **Latin** *fluxus* meaning “flow”
- A chemical cleaning agent, flowing agent, or purifying agent
- May have more than one function in a single welding process
- Used in both extractive metallurgy and metal joining

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Arc Welding

- Uses a welding power supply to create an electric arc between an electrode and the base material to melt the metals at the welding point
- Can use either direct (DC) or alternating (AC) current
- Can use consumable or non-consumable electrodes
- The welding region is usually protected by some type of shielding gas, vapor, and/or slag

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Oxyfuel Gas Welding

- Often called oxyacetylene welding, oxy welding, gas welding
- A welding torch is used to weld metals by heating them a temperature that produces a shared pool of molten metal
- The molten pool is generally supplied with additional metal called filler
- Filler material depends upon the metals to be welded.

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Regulators & Caps

- Regulators required to control fuel flow and pressure
- Need to be free from damage and defects
- Clean – no oil
- Install flashback / blowback arresters

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Gas Hoses & Fixtures

- Keep fixtures and hoses clean and in good repair
- Inspect hoses and fixtures prior to use
- Oxygen hoses are normally **GREEN** in color
- Fuel gas hoses (acetylene) are usually **RED**

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Solid State Welding

A welding process, in which two work pieces are joined under a pressure providing an intimate contact between them and at a temperature essentially below the melting point of the parent material

Bonding of the materials is a result of diffusion of their interface atoms

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Resistance Welding

Refers to a group of welding processes such as spot and seam welding that form the weld by heat generated by the electrical resistance of material vs. the time and the force used to hold the materials together during welding.

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Soldering

A process in which two or more metal items are joined together by melting and flowing a filler metal (solder) into the joint, the filler metal having a lower melting point than the work piece. Soldering differs from welding in that soldering does not involve melting the work pieces.

Below 450°C

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Brazing

- A metal-joining process whereby a filler metal is heated above melting point and distributed between two or more close-fitting parts by capillary action (getting into narrow spaces).
- The filler metal is brought slightly above its melting temperature while protected by a suitable atmosphere, usually a flux. It then flows over the base metal (known as wetting) and is then cooled to join the work pieces together.
- It is similar to soldering, except the temperatures used to melt the filler metal are higher (over 450°C)

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Other – Electron Beam Welding

- A fusion welding process in which a beam of high-velocity electrons is applied to the materials being joined
- The work pieces melt as the kinetic energy of the electrons is transformed into heat upon impact, and the filler metal, if used, also melts to form part of the weld
- Often done in conditions of a vacuum to prevent dispersion of the electron beam

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Other – Explosion Welding

- Do not try this at home
- Uses an explosive force to join metals
- Not a common welding practice
- Preparation of metals, types of metals, and size/nature of charge are critical elements

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Main Welding Hazards

- Toxic metal fumes from the welding process
- Fire and ignition hazards
- Burns – direct and slag
- Non-ionizing radiation
- Noise
- Materials used
- Equipment failure

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Toxic Fumes

Can produce one or more of the following health effects

- Chemical pneumonitis (inflammation of the lungs)
- Pulmonary edema (swelling of lungs and accumulation of fluid)
- Emphysema (loss of lung elasticity)
- Chronic bronchitis

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Primary Pulmonary Gases

- Ozone
  - From electrical arcs or ultraviolet photochemical reactions
- Oxides of Nitrogen
- Phosgene
  - From metals cleaned with chlorinated hydrocarbons
- Phosphine
  - From steel coated with phosphate rust proofing

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Non-pulmonary Gases

- Carbon Monoxide
  - Asphyxiate
  - Real problem in poorly ventilated areas
- Carbon Dioxide
  - Asphyxiate
  - Not normally listed as a toxic gas – is an issue in enclosed spaces

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Particulate Matter

- Aluminosis
  - Aluminum, aluminum oxide, aluminum hydrate
- Anthracosis
  - Black lung pigmentation
  - Deposit of carbon particles
- Siderosis
  - Iron oxide dust in lungs
- Stannosis
  - Tin oxide dust over prolonged period of time

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Pulmonary Irritants & Toxins

- Beryllium
- Cadmium
- Chromium
- Copper
- Fluoride
- Lead
- Magnesium
- Manganese
- Mercury
- Molybdenum
- Nickel
- Titanium
- Vanadium
- Zinc

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Pulmonary Irritants & Toxins

- Cadmium
  - Also impacts liver, kidney and/or bone marrow
  - Single exposure can be fatal
- Chromium
  - Particles mix with water vapor to from chromic acid
  - Hexavalent chromium compounds appear to increase chance of lung cancer

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Pulmonary Irritants & Toxins

- Copper
  - Metal fume fever
- Fluoride
  - Burns eyes and skin on contact
  - Excretion lags ingestion – builds up in bones
- Magnesium
  - Metal fume fever

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Pulmonary Irritants & Toxins

- Manganese
  - Highly toxic – can produce total disability after short exposure (few months)
- Nickel
  - Associated with lung and sinus cancers
- Zinc
  - Metal fume fever, zinc chills, brass chills
  - Short-term immunity can be built up (p. 836)

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Other Hazards

- Electrical contact
- Compressed gases
- Confined or restricted spaces
- Equipment failure and/or misuse

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Ventilation Systems

- General ventilation systems
- Localized ventilation systems
- Worker-specific ventilators

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Shields and Barriers

- For sparks and slag
- For ultraviolet radiation
- To control access to a working area
- To protect the welding process from outside interruptions

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Specialized PPE

- Masks and goggles
- Respiratory protection
- Gloves
- Shirts and leggings
- Aprons
- Boots
- Hearing protection

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Ignition Control

- Location of operations
- Venting/purging vessels or spaces
- Housekeeping
- Hazard assessments
- Fire watches and detection
- Special suppression systems

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